

Hitaansh Jain

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Education

New York University

GPA: 3.93/4.0 | BA, Computer Science, Minor: Mathematics

New York, NY

Expected May 2027

- Courses: Algorithm Problem Solving, Computer Organization, Linear Algebra, Artificial Intelligence, Software Engineering

Professional Experience

Header (joinheader.com)

Seattle, WA

Software Engineer Intern

May 2026 – Present

- Shipped customizable daily briefings to production: users reorder, toggle, and set detail level per section, backed by a JSONB layout model that assembles the LLM prompt dynamically instead of from hardcoded sections, plus a second-pass LLM-generated TL;DR.
- Owned Link Bankruptcy (bulk-closes a user's open tabs and returns a briefing summarizing them) end-to-end: built a paginated, authenticated FastAPI history endpoint with a React Native screen, and preserved submission metadata a scheduled cleanup job would otherwise have deleted.
- Wrote 400+ automated tests on a test-per-feature workflow across a multi-provider LLM pipeline, FastAPI/PostgreSQL backend, and React Native/Expo frontend, with every change merged to production through senior engineer review.

MathGPT (mathgpt.ai)

Remote

AI Intern (Part-Time)

May 2026 – Present

- Cut comparable run costs 39% and runtime by 50% via per-stage reasoning budgets and prompt caching, on a five-stage TypeScript/Next.js pipeline turning a calculus problem into a compiled LaTeX case study, flashcards, and a solution walkthrough.
- Designed independent verification for generated math: a critic model re-solves every problem in a clean context before any content is drafted, halting the pipeline on mismatch so unverified math never ships, enforced by per-stage Zod contracts with retries on validation errors.
- Owned the corpus extraction system (195 learning objectives) and a 6-table MySQL cache, covered by 210 automated tests including negative controls, on a three-intern Agile team.

Vardhman Infotech

Remote

Software Engineer Intern (Part-Time)

December 2024 – May 2025

- Migrated a Python inventory system serving 1,500+ retail locations from MySQL to Access to meet the client's zero-infrastructure requirement, refactoring a 73-function database layer into a metadata-driven architecture across 394 columns, 19 tables, and 700+ queries, delivered on schedule.

Projects

Stock Analyzer

April 2026 – June 2026

- Built and deployed a 3-tier stock analysis platform (Docker Compose, Flask, MongoDB) with rate-limit-aware retries for the OpenAI and SEC APIs, an 8-worker job pipeline, and 80% coverage gates in GitHub Actions CI/CD.
- Modeled MongoDB data across 6 collections with upserts ensuring zero data loss across concurrent jobs, and built session-based auth plus a polling job monitor with a live log view.

VocabLearn

February 2026 – May 2026

- Led backend on a 5-person Agile team, delivering a JWT-authenticated Node.js/Express REST API across 13 endpoints spanning auth, search, quiz, and CRUD, plus Gemini 2.5 Flash with structured JSON output for reverse-dictionary search-by-meaning.
- Engineered a quiz engine with distractor sets drawn from saved definitions, eliminating score double-counting via dual-layer deduplication (server-side Map plus client sessionStorage), and automated releases through a 3-workflow GitHub Actions pipeline.

Local RAG Search Platform

July 2025 – December 2025

- Built a fully air-gapped RAG system (FastAPI, Ollama, ChromaDB, React) for on-device semantic search over private documents, stamping source and page provenance into every chunk so citations survive generation.
- Diagnosed ranking flaws in a hybrid vector-plus-keyword retrieval pipeline with a 50-query ground-truth benchmark over 1,070 pages of zoning ordinances, motivating a BM25 redesign of keyword scoring.
- Implemented real-time token streaming to a React UI via the browser Streams API (ReadableStream, TextDecoder), eliminating UI blocking on full responses.

Swordfight (Unity, C#) | Gameplay & Systems Programmer

September 2023 – May 2024

- Shipped a 14-character top-down fighting game to itch.io on a 10-person student team, drawing 150k page views and 4.5 stars across 111 ratings.
- Owned the polymorphic weapon architecture and HFSM enemy AI behind multi-stage boss fights, and ran the PlasticSCM branch strategy that kept unmergeable Unity scenes conflict-free.

Computer Skills

Languages & Frameworks: Python (FastAPI, Flask), TypeScript (Next.js, React), JavaScript (Node.js, Express, React Native), Java, C#, SQL (PostgreSQL, MySQL), MongoDB

AI & Data Engineering: LLM pipelines (OpenAI API, Ollama, generator/critic verification), RAG (Retrieval-Augmented Generation), LangChain, ChromaDB, Pandas

DevOps & Workflow: Docker, Git, Claude Code, GitHub Actions, Render, CI/CD, Pytest, Vitest, Mocha, Agile/Scrum